

```
/*
```

```
Roll number : CSB24065
```

```
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```

```
*/
```

```
#include <stdio.h>
```

```
#include <time.h>
```

```
// O(1) Constant Time
```

```
void constantTime() {
```

```
    int x = 10;
```

```
    x = x * 2;
```

```
}
```

```
// O(n) Linear Search
```

```
void linearSearch(int arr[], int n, int key) {
```

```
    for (int i = 0; i < n; i++) {
```

```
        if (arr[i] == key)
```

```
            return;
```

```
    }
```

```
}
```

```
// O(n^2) Bubble Sort
```

```
void bubbleSort(int arr[], int n) {
```

```
    int temp;
```

```
    for (int i = 0; i < n - 1; i++) {
```

```
        for (int j = 0; j < n - i - 1; j++) {
```

```
            if (arr[j] > arr[j + 1]) {
```

```
                temp = arr[j];
```

```
                arr[j] = arr[j + 1];
```

```
                arr[j + 1] = temp;
```

```
            }
```

```
    }  
}  
}
```

```
int main() {  
    int n;  
    clock_t start, end;  
    double time_taken;  
  
    printf("Enter input size (n): ");  
    scanf("%d", &n);  
  
    int arr[n];  
  
    for (int i = 0; i < n; i++)  
        arr[i] = n - i;  
  
    // Constant Time  
    start = clock();  
    constantTime();  
    end = clock();  
    time_taken = (double)(end - start) / CLOCKS_PER_SEC;  
    printf("O(1) Constant Time: %f seconds\n", time_taken);  
  
    // Linear time  
    start = clock();  
    linearSearch(arr, n, -1); // key not present (worst case)  
    end = clock();  
    time_taken = (double)(end - start) / CLOCKS_PER_SEC;  
    printf("O(n) Linear Search Time: %f seconds\n", time_taken);  
}
```

```
// Quadratic time

start = clock();

bubbleSort(arr, n);

end = clock();

time_taken = (double)(end - start) / CLOCKS_PER_SEC;

printf("O(n^2) Bubble Sort Time: %f seconds\n", time_taken);


return 0;

}
```